Progress Report

- Increment 2 -
Group 14

1) Team Members

Samantha Bui

- sb21bn
- samanthabui

Ludginie Dorval

- lld22
- Ludginie

Antonio Garriga

- jaf21m
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William Lee

- wl23f
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Lillian Malik

- lam22g
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2) Project Title and Description

ManuFactor is a software solution designed to streamline accounting processes for manufacturing companies. It provides tools for Cost-Volume-Profit (CVP) analysis and budgeting for production (BP). Admin users can list/add users, and list/add data. Users can view and modify data. Viewers can only list the data. The system integrates database management, role-based user access, and secure authentication. By providing a straightforward tool for handling essential accounting tasks, *ManuFactor* enhances efficiency and accessibility in manufacturing operations.

3) Accomplishments and overall project status during this increment

Successfully finalized the integration between the front-end and back-end components of the project, ensuring seamless communication and functionality. Additionally, all pending HTML files have been completed. The team also addressed and resolved several SQL-related issues.

This increment we were able to add the functionality of budgeting production to the app through the product_bp.py file. This allows for another axis of analysis for a manufacturing company that would use our app to find out what they need in terms of production and direct material purchases based on the factors required to be entered on the front end. Update functions were also added to both product_cvp.py and product_bp.py to supply the front end with an easier time when it comes to their add and view data pages by eliminating the need for them to write one themselves.

The encryption and RBAC function as intended — we're able to log in as the initialized Admin, User, and Viewer, confirming that role-based access is working correctly. We've committed the Flask app routes for the login, results, home page, and Admin, User, and Viewer pages. The login, results, and home page HTMLs are complete and functional; HTML templates for the Admin, User, and Viewer pages have been set up; the next step involves implementing back-end functions and ensuring seamless integration with the front-end using these HTMLs.

4) **Challenges, changes in the plan and scope of the project and things that went wrong during this increment**

   a) Challenge connecting flask with database. Before Resolution: Flask had accurate functions, and the database had accurate functions. However, they didn't run properly due to a slight conflict in the connection. After checking, we concluded the code was accurate independently and shouldn't be called twice — just once. The connection issue needed to be resolved. After Resolution: After addressing the connection issue, Flask and the database are both working accurately, and the login system now functions without any errors. The implementation of encryption and RBAC is successful, allowing proper login access for Admin, User, and Viewer roles. This was a significant milestone, and the resolution has been committed and closed on GitHub.

5) **Team Member Contribution for this increment**

Samantha Bui

- **a) The Progress Report:**
  - a. **Project Title and Description:** Contributed to the revised project description that is the ReadMe on GitHub.
  - b. **Accomplishments and overall project status during this increment:** "The encryption and RBAC function as intended — we're able to log in as the initialized Admin, User, and Viewer, confirming that role-based access is working correctly. We've committed the Flask app routes for the login, results, home page, and Admin, User, and Viewer pages. The login, results, and home page HTMLs are complete and functional; HTML templates for the Admin, User, and Viewer pages have been set up; the next step involves implementing back-end functions and ensuring seamless integration with the front-end using these HTMLs."
  - c. **Challenges, changes in the plan and scope of the project and things that went wrong during this increment:** "Challange connecting flask with database. Before Resolution: Flask had accurate functions, and the database had accurate functions. However, they didn't run properly due to a slight conflict in the connection. After checking, we concluded the code was accurate independently and shouldn't be called twice — just once. The connection issue needed to be resolved. After Resolution: After addressing the connection issue, Flask and the database are both working accurately, and the login system now functions without any errors. The implementation of encryption and RBAC is successful, allowing proper login access for Admin, User, and Viewer roles. This was a significant milestone, and the resolution has been committed and closed on GitHub."
  - d. **Team Member Contribution for this increment:** Samantha Bui.

- **b) The Requirements and Design document:**
  - a. **Functional Requirements:**
    - i. Added: "User Management (High): refers to how different roles in the system (Admin, User, Viewer) manage users within the application. The Admin role is responsible for overseeing all aspects of user management, which includes creating, reading, updating, and deleting user accounts. The User role is limited to creating, reading, updating, and deleting specific data, while the Viewer role is limited to reading data only without modification capabilities."
    - ii. Added: "Product Management (High): refers to how different roles manage product data within the system. The Admin role holds full control over product management and is responsible for creating, reading, updating, and deleting product data, while the User role

is responsible for creating, reading, updating, and deleting product data. Viewers have a read-only role for product data."

- **c)** T I**mplementation and Testing Document**:
  - a. **Testing:** "We have communicated through issues to ensure accuracy and consistency in integration of the front-end code as well as the back-end code synchronously through in person meetings and asynchronously to ensure we are making progress. We have committed code and closed issued on GitHub."

- **d)** **The Source Code:**
  - a. Provided the Flask App "app.py" which implements Encryption and RBAC for Admin, User, Viewer.
  - b. Provided working HTML for login, results, home page and set up HTML for the Admin, User, Viewer pages. We will work on these HTMLs to display function values.

- **e)** **Video or Presentation:** Participated in video presentation with group.

Ludginie Dorval

- a) **The Progress Report:**
  For this increment, we successfully connected our HTML templates with our RBAC login system, which marks a major milestone in our project.
- b) **The Requirements and Design Document:**
  Helped define key non-functional requirements, including error handling and portability.
- c) **The Implementation and Testing Document:**
  Outlined our non-execution-based testing methods and described how we verified functionality through code review and manual validation.
- d) **The Source Code:**
  Ensured that the database was correctly initialized and functional, verified that the RBAC login directed users to the appropriate pages based on their role, and integrated the HTML templates with the Python backend.
- e) **The Video or Presentation:**
  I discussed my contributions during this increment and explained my role moving forward in the development process.

Antonio Garriga

- **a)** the **progress report**:
  - a. This increment we were able to add the functionality of budgeting production to the app through the product_bp.py file. This allows for another axis of analysis for a manufacturing company that would use our app to find out what they need in terms of production and direct material purchases based on the factors required to be entered on the front end. Update functions were also added to both product_cvp.py and product_bp.py to supply the front end with an easier time when it comes to their add and view data pages by eliminating the need for them to write one themselves.
- **b)** the **requirements and design document**:
  - **a. Provided Lilian with all the necessary information on what would be required for the database in regards to product_bp.py to inform the updates to the use case diagrams**
- **c)** the **implementation and testing document**:
  - **a. Execution based functional testing of both product_cvp.py and product_bp.py to ensure that they both do exactly what they should do logically, such that it is a non-issue when combining it with the frontend**

- **d)** the **source code**:
    - a. Created the product_bp.py file consisting of the following functions:
        - i. Get and Updates to pull and push data to and from the database
        - ii. Logical functions to calculate the required production and required purchases
        - iii. A main function to tie it all together and return a cohesive json format result to the front end
    - b. Updated the init_database.sql file to now have a table for product_bp
- **e)** the **video or presentation:**
    - a. Talked specifically about my contributions in terms of writing the source code I mentioned just above. Also talked about the goals of helping Sam, Ludgi, and Lilian in Increment 3 to ensure that the connection between the backend logic and the frontend comes together seamlessly

William Lee

- **a)** the **progress report**:
- **b)** the **requirements and design document**:
- **c)** the **implementation and testing document**:
- **d)** the **source code**:
- **e)** the **video or presentation:**

Lillian Malik

- **a)** **the progress report**:
  Revised and updated the information for Increment 2 in Sections 3, 6, and 7, while also making the necessary contributions to Section 5.
- **b)** **the requirements and design document**:
  Revised and updated the Case, Class, and Sequence diagrams, as well as updated the information in all sections, excluding Section 3, from the previous increment to reflect our current progress.
- **c)** **the implementation and testing document**:
  Minimal changes were required; however, I updated the non-execution-based testing (Section 5) to reflect my role as the de facto SCRUM master, which allows me to explain this section most effectively.
- **d)** **the source code**:
  Updated the list_users, add_users, list_data, and add_data HTML files, while also adding the cvp_results and bp_results pages to provide a more detailed and thorough analysis of the data.
- **e)** **the video or presentation:**
  Facilitated the recording of the presentation, and provided an in-depth overview of the project, as well as my role in this Increment.

6) Plans for the next increment

For the upcoming increment, our primary focus will be on finalizing any remaining features and ensuring all functionality is fully implemented. We will enhance the user experience by incorporating CSS and JavaScript to improve the overall design, interactivity, and ease of use. Additionally, we will focus on refining the project to ensure seamless user access and greater control over their account settings, addressing any outstanding issues, and delivering a polished, user-friendly final product.

7) Stakeholder Communication

Subject: Progress Update on ManuFactor – Increment 2

Dear Stakeholder,

I am writing to provide an update on the progress of ManuFactor, our web-based accounting platform.

Current Status:
In this increment, we focused on testing the functionality of the code and finalizing the HTML files. Due to the shorter, three-week timeframe, our efforts were more limited compared to the previous increment, where we established encryption and database management.

Challenges:
The limited time available required us to prioritize essential tasks, resulting in fewer meetings than usual. Additionally, we encountered some challenges with SQL and database connection, which slowed down our progress in certain areas. Despite these setbacks, we successfully made significant progress within the given timeframe.

Next Steps:
For the upcoming increment, our primary focus will be on finalizing any remaining features and ensuring all functionality is fully implemented. We will enhance the user experience by incorporating CSS and JavaScript to improve the overall design, interactivity, and ease of use. Additionally, we will focus on refining the project to ensure seamless user access and greater control over their account settings, addressing any outstanding issues, and delivering a polished, user-friendly final product.

Thank you for your continued support. Should you have any questions, please do not hesitate to reach out.

Best regards,
Group 14, ManuFactor

8) Link to video
https://www.youtube.com/watch?v=d2__wfaCBkQ